

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 325 — Past Questions & Materials

by Renance

Course: MAR 325

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 300

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Paper 1 of 3 — MAR 325: MARINE ENGINEERING MATERIAL EXAMINATION: Second Semester 2020 2021

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN DELTA STATE
MAR 325: MARINE ENGINEERING MATERIAL EXAMINATION:
300 LEVEL SECOND SEMESTER
TIME 2 HOURS.

Instruction: Answer any 4 questions.

Question 1

Iron ore, coke and limestone are charged to the furnace through the double bell-and-cone gas-trap system for the production of pig iron.

- a. What are two chemical formulas of the ore?
- b. Does pig iron has any engineering value?
- c. The coke is oxidized to what?
- d. Are there other ways to process iron ore without a blast furnace?
- e. What is the use of carbon monoxide in the furnace? (25 marks)

Question 2

- a. What makes steel different from pig iron?
- b. Describe the phase equilibrium of an Austenitic steel.
- c. How does Martensitic steel formed?
- f. How does one modify the steel to improve toughness? (25 marks)

Question 3

- a. Describe effects of different alloying elements of Nickel, Titanium and Chromium
- b. What is the purpose of adding alloying elements?
- g. Describe how microstructure of steel relates to its use
- h. Describe at least 5 important plastic materials relevant to Marine Engineering you have learnt in this course. (25 marks)

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Question 4:

- a. Describe the various heat treatment of steel
- b. What are the benefits of plastic materials over metals? (25 marks)

Question 5:

- i. What are complex ferrous alloys?
- ii. What are the features of High Speed Steel as a complex ferrous alloy?
- iii. Describe the differences and similarities between welding, soldering and bracing?

(25 marks)

Question 6

- a. What are five benefits of Aluminum over iron?
- b. Aluminum is prone to porosity due to bubbles of Hydrogen. How?
- c. How can the porosity issues in Al castings be treated?
- d. Apart from Al and its alloys, what other non-ferrous metals that are relevant to us in Marine Engineering? Where can we use them?

(25 marks)

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Paper 2 of 3 — MAR 325 MARINE CORROSION (MAR 325) 2020 2021

Level 300 • Second Semester • PastQuestion

Question 4.

i necessary.

Desmostrate with diagrams and equations where

(10 marks

i. How is corrosion measured? ita)

. 9

ii, Why do materials corrode? _ ; 5

iti fity is Iron (Fe) said to be unprotective in water environment? (5 marks)

4, a. Discuss the two methods of corrosion testing O inaris)

b, Discuss the concept of corrosion inhibition oo (5 marks)

c. Discuss different types and applications of corrosion inhibitors

(10 marks)

Question 5.

(a). What are the two conditions to be met wi

hile designing for protection using

anodes?

(5 marks)

(b). Why is Aluminium (AD said to be protective? (5 marks)

@. What are the major seawater Parameters affecting Cathodic Protection in-Site

(10 marks)

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN.

DEPARTMENT OF MARINE ENGINEERING.

SECOND SEMESTER EXAMINATION, 2020/2021 SESSION.

COURSE: MARINE CORROSION (MAR 325)

Instruction: Answer question | and any other 3. questions.

Time allowed: 1.5 hours.

Question 1.

You have been consulted by a Marine company to design and install magnesium anodes on a rectangular barge. The details supplied are as follows: length overall = 50m, beam = 12m. Thickness of steel plete used = 2mm. Take steel density as 7900kg/m'. You may also conside, that the barge can carry Ston of weight.

Take: $p = 500\text{ohm-cm}$ $I_c = 0.1\text{mA/m}^2$ $E_c = -0.670\text{v}$ and coating breakdown

=1

Mg: 5 x 10cm, $E_{mg} = -1.7\text{mv}$; (40 marks)

Question 2.

- a. As the material engineer on site, you have been presented with the option of choosing between a corrosive material and non-corrosive material for the construction of the ROV system to be used for the inspection of a structure. Discuss what will influence your decision. (10 marks)
- b. What is corrosion? (5 marks)
- c. Can corrosion be stopped? (5 marks) |

Question 3.

- i. list and explain 5 types of corrosion you know (10 marks)
- ii. Describe different types of corrosion inhibitors? (5 marks)
- iii. Is it proper to install anodes above the submerged part on a floating structure? Discuss. (5 marks)

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Paper 3 of 3 — MARINE CORROSION ENGINEERING (MAR 325)

Level 300 • Second Semester • Handout

MARINE CORROSION ENGINEERING (MAR 325)

A) Fundamentals of corrosion

- Corrosion principles and types of corrosion
- General corrosion (rusting)
- Localized corrosion (pitting, fretting, cavitation, crevice filiform)
- Galvanic corrosion
- Environmental corrosion cracking (Hydrogen induced cracking, Stress corrosion cracking)
- Flow-assisted corrosion (Erosion, corrosion, impingement cavitation)
- Inter granular corrosion
- High-Temperature corrosion
- Microbial corrosion
- *Properties of seawater
- *Metallurgical factors influencing corrosion
- *Pourbaix diagram

B) Corrosion control and monitoring

- Internal corrosion
- Measurement and monitoring of internal corrosion
- Coatings and inhibition (protective coating for ships, underwater antifouling paint, pipeline coating)
- Material choice (Corrosion resistant alloys, stainless steel)
- Cleaning of steel work (Acid pickling, dry & wet blast cleaning)
- Cathode protection of offshore structures, pipeline risers and associated equipment (pipeline survey and inspecting sacrificial anode and impress current system anode materials)

Fundamentals of Corrosion

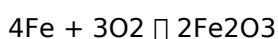
Corrosion is the gradual destruction of material, usually metals due to a chemical or electrochemical reaction with their environment. This reaction can cause the material to weaken or disintegrate, leading to structural failure.

Corrosion rate is measured in units like mils per year (mpy), millimeters per year (mm/year) or inches per year (ipy) it can also be weight loss per unit area

Reactions involved in metal corrosion

Reaction1:Oxidation

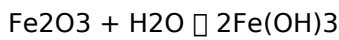
metal (iron) + oxygen → Iron oxide (rust)



The metal surface reacts with oxygen to form a thin layer of iron oxide (rust).

Reaction 2: Moisture ingress

Iron oxide (rust) + water → Iron Hydroxide



Moisture enters the cracks and reacts with the iron oxide to form iron hydroxide

Reaction 3: Acid Formation

Iron Hydroxide + Carbon Dioxide (from air) → Iron Carbonate (acidic)



The iron hydroxide reacts with carbon dioxide to form an acidic iron carbonate

Reaction 4: Corrosion Acceleration

Iron carbonate (acidic) + metal (iron) → more iron oxide (rust) + Hydrogen gas



The acidic iron carbonate accelerates the corrosion reaction, producing more iron oxide (rust) and hydrogen gas.

Reaction 5: Debris Accumulation

Iron oxide + other corrosion products → Debris accumulation

Fe_2O_3 + other corrosion products

The corrosion products like rust and iron carbonate, accumulate in the contact area, further accelerating the corrosion process.

Types of corrosion

1. General corrosion (rusting)

General corrosion is commonly recognized as rusting of iron. This is when metal slowly rust away due to exposure to a corrosive environment, like sea water or air. Water and air (containing oxygen) meets on the surface of the metal. The oxygen reacts with the iron to form a compound called iron oxide (rust). The moisture helps create an electrochemical reaction, where the iron acts as an anode and the oxygen acts as a cathode. As the reaction continues, the iron oxide (rust) grows and spreads, weakening the metal.

2. Localized corrosion

Localized corrosion occurs on specific sites on metal surfaces and in specific environmental conditions. Corrosion activities on this specific sites may start and stop with a change on the environment. In localized corrosion, the rate of corrosion on the same material is greater in some areas than others.

3. Pitting corrosion

Pitting is a type of corrosion that causes deep, narrow holes in metal. It is a deep narrow corrosive attack which often causes rapid penetration of the substrate. Using an offshore oil plant as an example, overtime tiny scratches or floors can appear on the surface of the metal legs of the platform. This scratch creates a tiny hole where corrosive seawater can seep in and cause damage. As the corrosion starts, a tiny "Battery" forms inside the scratch, with the scratch (anode) corroding and the surrounding metal (Cathode) not corroding. Corrosive acids and salt build up inside the scratch, making it worse.

Figure 1: Perforation of tube with millscale

The corrosion process happens in stages;

- Initiation: The scratch appears and corrosion starts.
- Propagation: Corrosion spreads deeper (Deep pitting)
- Termination: Corrosion stops if the scratch is fixed or the environment changes
- Reinitiating: Corrosion restarts if the scratch is exposed again

4. Fretting corrosion:

Fretting is a localized corrosion phenomenon that is caused by mechanical stresses. It is as a result of slight relative motion (for example as in vibration) of two substances which at least one of them is a metal, that are in contact. (oxide layer disrupts, moisture enters the cracks, the moisture reacts with metal causing corrosion leading to the formation of corrosive products like rust or oxide).

Figure 2: Fretting corrosion

Effects of fretting corrosion;

- Pits form on metal surface (It can lead to structural weakness, pits can also act as stress concentrators, increasing the risk of cracking).
- Galling (Roughening of surfaces can cause wear on adjacent parts).
- Fatigue (Cracking of materials)
- Seizing (Freezing of moving parts)

5. Crevice corrosion:

Let's say we have two metal pipes connected by bolt. If the bolt is not tightened properly, a small gap can form between the pipes. Seawater can enter this gap and cause crevice corrosion. In other words crevice corrosion occurs when seawater or substances get stuck in the crevice (Small gaps in metal parts), reacting with the metal thereby causing corrosion. The corrosion process creates more substances that build up in the crevice. The buildup of these substances slows down the corrosion process, but doesn't stop it.

Crevice corrosion happens when there's a small gap or crevice between two metal and a non-metal. This gap restricts the flow of air and water around the areas, creating a unique environmental that can lead to corrosion.

There are two main reasons why crevice corrosion occurs;

- Oxygen concentration cells: This happens when there is a difference in oxygen levels between the service and the surrounding area. This difference creates an electric potential that drives the corrosion reaction.
- Metal ion concentration cells: This happens when there's a buildup of metal ions in the crevice, which creates an electric potential that drives the corrosion reaction.

6. Cavitation:

Cavitation is a type of damage that happens when vapor bubbles form and collapse

on metal surfaces in contact with moving water. A ship propeller is like a fan spinning through seawater. As it moves, it creates bubbles that collapse and damage the propeller surface, leading to cavitation erosion.

7. Filiform:

Filiform corrosion is a type of corrosion that happens under coatings or metal surfaces in offshore settings. It looks like a network of tin, random threads or filaments of corrosion products. Imagine a coated metal pipe on an offshore platform. If the coating is damaged or degraded, moisture can seep underneath and create a specific oxygen cell. This cell causes corrosion to advance in a tin filament, about 0.1mm wide, under the coating.

How it works;

- Moisture seeps under the coating, creating a small area with low oxygen levels (called an anode).
- The surrounding area with higher oxygen level becomes the cathode.
- The difference in oxygen levels creates an electric potential difference between the anode and cathode.
- This electric potential difference drives the corrosion reaction, causing the metal to dissolve at the anode.
- The corrosion products form a tin filament that advances under the coating, creating the network of threads.

8. Galvanic corrosion:

Galvanic corrosion happens when two different metals or a metal and a conductive non-metal are connected in a corrosive environment, like seawater. This connection creates a tiny electric circuit that speeds up corrosion. Example, think of a ship's hull made of iron, connected to a copper propeller in seawater. The copper propeller becomes the cathode (Less active metal) while the iron hull becomes the anode. The connection creates a galvanic circuit, causing the iron hull to corrode faster.

9. Environmental corrosion cracking:

This type of cracking is caused by the combined action of corrosion and tensile stress.

- Hydrogen – Induced Cracking (HIC); This is a type of corrosion that causes brittle failure in metals, especially high-strength alloys. It happens when atomic hydrogen enters the metal surface and gets trapped inside. Atomic hydrogen is produced on metal surface through processes like welding. This hydrogen can then diffuse into metals lattice structure, causing it to crack.
- Stress corrosion cracking; It is a type of corrosion that causes sudden brittle failure in metals (breaks or shatters easily), especially pipelines. It happens when a metal is under tensile stress and exposed to a corrosive environment.

Figure 3: Stress corrosion cracking

There are 2 main types:

- High-PH SCC (Alkaline environment, PH 9-13)

□ Near-Neutral-PH SCC (Mildly acidic environment, PH 5-7)

Factors involved in stress corrosion cracking

The main factors to be considered are the environmental conditions, the stress conditions and the alloy under consideration.

- i. Environmental conditions: For most constructional alloys, stress corrosion cracking occurs only in conditions specific to that particular alloy. As already noted, austenitic stainless steels are susceptible in chloride-containing environments. This does not mean that such steels will always crack if used under marine conditions. The concentration and temperature of the chloride containing solutions and the stress level all determine the likelihood of such attack, so it is only under a certain set of conditions that problems may arise. Because of the combination of conditions it is not always a straightforward matter to determine the probability of attack, particularly in view of the importance of the local environment at the surface. The classic example of stress corrosion cracking of riveted mild steel boilers (caustic embrittlement) occurred even in situations where the alkaline water contained less than the concentration of sodium hydroxide considered necessary to promote cracking. This was because of the evaporation of water from the seams leading to concentration of sodium hydroxide, which led to stress corrosion cracking. Similar situations have arisen with stainless steel back-boilers using domestic water where chlorides have concentrated. This is also an illustration of the problems that can be accentuated under heat transfer conditions. It is not possible to provide more than a general guide to the influence of environmental factors on the susceptibility of alloys to stress-corrosion. Most constructional alloys are susceptible under some sets of circumstances, but this does not mean that problems are likely to arise in more than a few cases. Unfortunately, it is not easy or in some cases even possible to predict the exact situations where cracking will occur. Apart from the chemical composition of the solution in contact with the alloy surface, temperature and, in some cases, the presence or otherwise of oxygen will affect the process. It has been reported that in sea-water Type 316 steels, particularly where the metal surface becomes encrusted with salt, may be susceptible to stress corrosion cracking at temperatures above 140°F (60°C).
- ii. Stress Stress: is clearly one of the most important factors in determining the probability and course of stress-corrosion cracking, but it is not an easy matter to predict stress levels at which failure will occur. This, in part, arises from the nature of the stresses that can cause cracking of alloys. They may be applied but frequently they are internal stresses arising from fabrication procedures such as welding and cold working.
- iii. Alloys: As the electrochemical characteristics of alloys influence their susceptibility to stress corrosion cracking, changes in alloy composition would be expected to have a marked effect on their performance in relation to stress corrosion. This is clearly demonstrated by the family of steels termed 'stainless' where increases of nickel content above about 8-10% in austenitic steels increases the resistance to stress corrosion—and with 40-45% nickel, the resistance is greatly enhanced. Heat treatments can also influence an alloy's susceptibilities in various ways, e.g. by affecting internal stresses. Further comments on susceptibility to

stress corrosion cracking is provided in the sections of the book dealing with specific alloys.

10. Flow- assisted corrosion:

Flow assisted corrosion occurs due to the combined action of corrosion and fluid flow.

They can be divided into three types;

□ Erosion corrosion

□ Impingement

□ Cavitation

□ Erosion corrosion:

This occurs when the velocity of the liquid is sufficient to remove protective films from the metal surface. The corrosion sites are often localized in areas where flow are aberrations and turbulences occur due to material discontinuities. Such discontinuities could be the weld or any notch in the metal. For corrosion to occur, the adhering of the protective film to the substrate is an important factor.

□ Impingement:

This is caused by turbulence or impingement of flow. Imagine a strong water jet hitting a metal surface, like a power washer. The force of the water, combined with air bubbles and particles in the seawater accelerate the corrosion.

Figure 4: Impingement attack on a (a) pipe and (b) a cylinder

11. High -Temperature Corrosion:

This is usually associated with the formation of oxide or sulphide scales. In some cases, the swelling of the wall of the material is also noted. This is a form of material degradation that occurs at elevated temperatures. In this type of corrosion, a metal that is exposed that high temperature reacts with oxygen, Sulphur, or other gases to form scales. The scales can flake off, exposing fresh metal to further corrosion.

12. Inter granular corrosion:

This is a preferential attack at or adjacent to grain boundaries of metals. Imagine a metal, like stainless steel, made up of tiny grains. The boundaries between these grains can be weak spots where corrosion can start. Corrosion occurs when impurities accumulates at grain boundaries making them to become weak and prone to corrosion. Corrosion attacks the grain boundaries, causing cracks.

Example; Consider an offshore oil rigs stainless steel pipes, which are welded together. If the welding process isn't done correctly, it can cause inter granular corrosion. The high temperatures during welding can create impurities at the grain boundries, making them weak. When the pipes are exposed to acidic environment, like seawater, intergranular corrosion can occur, leading to cracks.

Figure 5: Intergranular corrosion

13. Microbial corrosion:

Microbial corrosion occurs when tiny living things like bacteria and fungi stick to metal surfaces and create a slimy layer (biofilm). As they eat and grow, they produce

yucky stuff that can dissolve in metal. Additionally, the microorganisms can also break out protective layers on the metal surface, making it more susceptible to corrosion.

Properties of seawater

The concentration of dissolved materials in the sea varies greatly with location and time. The most important properties of seawater are;

- High salt concentration, mainly sodium chloride
- High electrical conductivity
- Relative high and constant PH
- Buffering capacity
- Solubility of gases, of which oxygen and carbon dioxide in particular are of importance in the context of corrosion.
- The presence of a myriad of organic compounds
- The existence of biological life, to be further micro fouling (e.g. Bacteria) and macro fouling (e.g. Seaweed, barnacles and many kinds of animals and fish)

Some of these factors are interrelated and depend on physical, chemical and biological variables such as depth, temperature, intensity of light and the availability of nutrient.

Metallurgical factors influencing corrosion

There are several metallurgical factors that influences corrosion such as;

- Allure composition: The presence and concentration of alluring elements, impurities and residual element can significantly impact corrosion resistance.
- Microstructure: The arrangement of grains, phases, and defects in the metals micro structure can affect corrosion behavior.
- Grain size and orientation: smaller grain sizes and specific grain orientations can improve corrosion resistance
- Surface finish and roughness: a smooth surface finish can reduce the risk of corrosion by minimizing crevices and areas for corrosion to initiate.
- Residual stresses and strain: Internal stresses and strain can increase the susceptibility of a metal to corrosion, particularly stress corrosion cracking.

Pourbaix diagram for iron

A pourbaix diagram, also known as an E-PH diagram, is a graphical representation of the thermodynamic stability of different species of a metal in aqueous solutions. It is a powerful tool for understanding corrosion and material stability.

Figure 6: Pourbaix diagram for iron

Axes;

X-axis (PH): represents the acidity or basicity of then solution

Y-axis (E): represents the electrochemical potential (voltage) of the solution

Regions

- Immunity: The metal is resistant to corrosion (lower left)
- Corrosion: the metal corrodes (upper left and lower right)
- Passivity: A thin oxide layer forms protecting the metal (upper right)

Pourbaix diagrams are specific to each metal and solution.

CORROSION CONTROL AND MONITORING

As corrosion results from a reaction between an alloy and its environment, the basic control methods are concerned with treating the environment or selecting suitable alloys. The methods can be grouped into four main categories, irrespective of which of the two factors— environment or material—are of major importance.

- (i) Use of coatings.
- (ii) Selection of materials that will resist a particular environment.
- (iii) Cathodic protection.
- (iv) Control or treatment of the environment.

Causes of internal corrosion

- Chemical reactions on internal surfaces
- Microbiological reactions
- Pressure and velocities of gases and liquids (pipelines and risers)
- Inclination of pipe lay

Factors affecting internal corrosion

- Flow phase and pipe lay
- Multiphase flow and inclination
- Slug velocity and frequency
- Water percentage in the pipe system

Corrosion Mitigation

- Internal coating
- Chemical injection (corrosion inhibitors)
- Selection of inhibitor types and concentration
- Monitoring and adjusting inhibitor levels

Data collection for internal corrosion assessment

- Fluid chemistry (free water, oxygen, hydrogen, Sulphur)
- Pipe material reactions with gases, liquids and solids
- Bacteria culture and operating stress level

Internal corrosion monitoring techniques

- Coupons
- Acoustic emission
- Real-time ultrasonic
- Hydrogen patches

The above listed monitoring techniques are grouped into;

- Intrusive methods

□ Non-intrusive methods

□ Intrusive methods: Usually involve inserting a coupon or probe into the pipeline or other topside equipment.

□ Non-intrusive methods: Do not interfere with operations

Corrosion coupon: A coupon is a small piece of metal (like a tiny rectangle) that is inserted into a pipeline or piping system. The coupon is made of the same material as the pipeline, so it represents the pipelines condition. The coupon is exposed to the same environment as the pipeline, including the fluids, gases, and temperatures, which makes it corrode over time just like the pipeline would. The coupon is removed and measured to see how much it has corroded. The corrosion rate on the coupon is compared to the expected corrosion rate to determine if the pipeline is corroding as expected.

Acoustic Emission monitoring: It's a way to detect corrosion inside a pipeline or piping system by listening to tiny sounds it makes. Small sensors are attached to the outside of the pipeline. The sensors listen for tiny sounds (like cracks) that are too quiet for humans to hear. When corrosion occurs, it makes tiny sounds as the metal breaks down. The sensors send the sound signals to a computer for analysis. If the computer detects a pattern of sounds that indicates corrosion, it triggers an alarm. It is a non-intrusive method, to detect corrosion

Real-time ultrasonic monitoring: A small ultrasonic probe is inserted into the pipeline. The probe sends a high frequency sound waves through the pipeline wall. The sound waves bounce off the internal surface of the pipe and return to the probe as echoes. The probe measures the time it takes for the echoes to return which indicates the thickness of the pipeline wall.

Hydrogen Patches: They are small sensors that detects hydrogen gas, which is often a sign of corrosion inside a pipeline or piping system. These small patches are attached to the outside of the pipeline, the patches contain a special material that reacts with hydrogen gas, which is produced when corrosion occurs. When hydrogen is detected, the patch changes color, indicating corrosion.

Coatings and Inhibition

Corrosion protection involves using various methods to prevent or slow down corrosion. The term coating points to a chemical compound, and a coating is essentially composed of synthetic resins or inorganic silicate polymers. When applied to a prepared surface, these chemicals will form a coating that resists the harsher environment of industry and sea. In terms of adhesion, toughness, and resistance to weather and seawater, these applications perform better than a paint system.

Coatings are the most common method by far, in terms of steel tonnage, for controlling corrosion. The basic concept is not usually electrochemical in nature, it is simply to insulate the alloy from the corrosive environment. In practice,

however, many problems arise and often insufficient attention is paid to them because of the apparently simple nature of coating systems. A range of different types of coating is available, the most common of which are paints although metal coatings, plastics, waxes, greases and other materials are also used. Organic coatings are basically of two types : those that are applied manually, e.g. by spraying or brushing, and those that are applied in works. All organic coatings are permeable to moisture and oxygen to a greater or lesser extent and to overcome this, there has been a growing tendency to specify thicker coatings for marine situations. This has been made possible by the nature of coatings such as epoxies and urethanes which harden or cure chemically. Such coatings are basically barrier coatings to the environment. Most paint films are not impermeable to water or oxygen. This ability of oxygen and moisture to diffuse through a paint film indicates that the protective action arises not through control of diffusion but rather to the high ionic resistance of paint films, which impede the migration of ions so reducing the corrosion current to a low value (resistance control). Apart from metallic pigments in organic (or inorganic) binders, metals are also used as coatings. They are basically insulating or isolating barriers to the environment. Furthermore, being metals, they corrode— albeit at a lower rate than steel. Although sprayed metal coatings are porous, even they to a considerable extent act as barrier coatings. The coatings widely used on steel, i.e. zinc, aluminium and cadmium, have potentials more negative than that of steel, i.e. they are anodic to steel so at discontinuities or damaged areas they 'protect' the steel. This is an additional advantage but they are primarily used because they corrode at a lower rate than does uncoated carbon steel. Other coating metals such as copper, nickel and chromium have corrosion potentials that are more positive than that of steel. This means that at discontinuities, the steel will corrode at an accelerated rate, so these cathodic coatings should be of a continuous nature with a minimum of defects. In practice, however, in many environments corrosion products tend to block small pores in the coating. Probably the most resistant coating in marine environments is vitreous enamel. The fact that such coatings are virtually never used for structures illustrates the essential elements in the choice of coatings. Apart from their corrosion resistance, they must be capable of easy application to a variety of sizes and shapes of sections and must be reasonably cheap, properties not possessed by vitreous enamel.

Types of coatings

- Organic coatings (e.g. paints, plastics, bitumen)
- Enamel coatings
- Metallic coatings
- Cement mortar coatings
- Organic coating: Refers to a layer of protection made from organic materials such as paints, varnishes, or polymers, applied to a surface to shield it from

corrosion, wear, and damage. It is usually applied using various techniques such as spraying, brushing, and dipping.

□ Enamel coatings: Enamel coating is a hard, glossy layer made from a mixture of glass powder (or ceramic material), metal oxides (for color and durability), and other additives (for specific properties). The mixture is applied to surface using various methods such as; spraying, dipping, and brushing. The coated surface is then heated in a kiln or oven to melt the glass powder, cooled to form a hard, glossy layer.

□ Metallic coatings: Metallic coatings are thin layers of metal, usually applied using various techniques such as electroplating (depositing metal ions into a surface using electricity) or spraying (applying molten metal onto a surface). Common types of metallic coatings are; Zinc (protects against corrosion), chromium, Nickel (corrosion resistance) and silver (antimicrobial properties)

□ Cement mortar coating: is a protective layer applied to surfaces to shield them from damage, corrosion, and deterioration. It is a mixture of cement (binding agent), sand (aggregate), water (mixing agent) and optional additives (for specific properties). It is done usually by spraying, pumping and troweling (with a tool).

Coating provide a barrier between the corrosive environment and the material to be protected.

PROPERTIES OF COATINGS

Coating have various properties, such as;

- Adhesion
- Chemical stability (Ability to resist degradation or reaction when exposed to certain chemicals).
- Mechanical resistance (ability to withstand physical forces, impacts & stresses).
- Permeability (ability to allow or prevent molecules to pass through).

IMPORTANCE OF COATING RESISTANCE

Coating resistance is a critical factor in determining the effectiveness of a coating. It's affected by factors like water absorption, temperature and the type of resin used (Ability of a coating to withstand and resist various types of stresses, environments and exposures without failing).

COMMON ISSUES WITH COATING

- Holidays (defects or pores) in coating
- Blistering (electrochemical reaction causing coating to lift off)

Offshore structure can, from the corrosion standpoint, be divided into three broad zones: submerged zone; splash zone; atmospheric zone.

Figure 7; Offshore structure: areas of corrosion and types of control

- i. Atmospheric zone: This is the above the splash zone that is not affected by seawater.

Corrosion control is through coating systems, no cathodic protection needed. The use of dissimilar metal joints or insulation should be avoided in this zone. This zone is not wetted or affected by the rise of tidal waves; hence, it is not in the electrolyte. The corrosion control of this zone is achieved through the application of a suitable coating system. Given that this zone is not in contact with the electrolyte (seawater), the coating does not need to be complimented by cathodic protection, the structure design essentially tries to minimize the exposed steel surface area in this zone. Minimizing the exposed surface area is primarily done through the use of tubular members. Clean welds with good profiles that blend into the parent metal then prevent any crevices that might promote corrosion, and the boxing in of the steel structures can accomplish the same goal. Substituting steel with corrosion resistant materials and nonmetals wherever possible is another example of designing a structure with a reduced tendency to corrode. Planners should avoid designs in which dissimilar metals are joined, however, or they should provide insulation between dissimilar metal joints.

ii. Splash zone: Intermittently wetted by seawater due to tides and wind. Corrosion protection requires a different approach than submerged zone. Splash and atmospheric zones do not become covered with marine fouling and so do not receive the same protection as other areas of piling. Furthermore, the splash zone is continually wetted and dried, so the corrosion rates tend to be highest in this area—from 100-250 $\mu\text{pi/y}$. The corrosion in the atmospheric zone depends upon the height above the splash zone and is within the range 50-100 $\mu\text{pi/y}$. The range of the zone varies with the height of the sea's rise and fall due to daily tide cycles in the specific geographic area. For example, the range of the splash zone in the Gulf of Mexico is about 6 ft., but it is about 33 ft. in North Sea and about 1 m (3.32 ft.) in the North China Sea. The corrosion protection of material in this zone requires a different approach than the conventional approach discussed for the immersed zone.

iii. Immersed/submerged zone: This zone is in constant contact with seawater. The corrosion protection is through high-performance coating system and cathodic protection (CP) system. The submerged zone is the section of the structure that is below the lowest end of the splash zone and is always below the sea level. From the cathodic protection point of view, this portion of the structure is always in the electrolyte. The corrosion protection of this zone is achieved through a well-designed program that includes the application of a high-performance coating system supplemented by an equally well designed CP system. To reduce corrosion risk, the structural design preferably uses tubular construction, as tight or recessed corners and crevices are difficult to protect in this zone. A stress-reduction design is also encouraged to minimize fatigue stress, and where required, the design should also relieve the stress placed on welded members. Structure-pilings, skirt-piles, and centralizers are considered to be part of the structure for CP system design. When these members are encased within the jacket legs, they are bonded to the legs by welding, and their surface area is to be considered while calculating the required polarization current. The parts that are not welded must be electrically bonded to the structure to ensure that they are part of the CP system. Although still in the submerged zone, members below the mudline generally experience a very low corrosion rate, as there is scant oxygen or turbulence to act on the

steel; however, members buried in the seafloor should be made part of the total CP system.

MATERIAL CHOICE

Generally, the approach to corrosion control is to consider carbon steel in conjunction with other control methods such as inhibitors, cathodic protection and coatings, and only to choose more resistant alloys where carbon steel would be uneconomic. Factors other than corrosion resistance may also play a role, e.g. forming, shaping, casting etc. Carbon steel is the cheapest of all constructional alloys. However, for a limited number of small components or for certain precision parts, material costs may be a minor consideration and alloys other than steel might be used. For large components and structural members, material costs are an important element in the overall economics and generally more resistant alloys would be used only in special circumstances. The use of corrosion resistant alloys can be considered as an increase in anodic control. This arises from either the formation on the surface of an alloy of protective films or, possibly, the sluggishness of the dissolution reaction. Metals that form these protective films include Cr, Al, Ti and Ni and alloys such as stainless steels, 'Monel' and 'Inconel' containing a high proportion of such metals as alloying elements also form such films. Sometimes thicker films which—unlike those discussed above—are visible may be formed, e.g. the patina of basic copper sulphate and basic copper salts formed on copper.

Corrosion resistant alloys

- i. Stainless steel
- ii. Titanium
- iii. Copper-Nickel alloys
- iv. Brass (copper-zinc)
- v. Bronze (Copper-Tin)

Stainless steel

- Stainless steels contain a minimum of 11-5% chromium, which provides corrosion resistance.
- There are 3 main categories of stainless steels namely martensitic, ferritic and austenitic each with different mechanical and corrosion resistant properties.
- Austenitic stainless steels are the most widely used for marine applications due to their high corrosion resistance.
- Velocity has a significant influence on corrosion performance in stainless steels, with higher velocities reducing corrosion.
- Nickel additive improve corrosion resistance particularly in reducing pitting and crevice corrosion.

CLEANING OF STEEL WORK

1. **ACID PICKLING:** Is a chemical cleaning process used to remove impurities, scale, and rust from steel surfaces. A solution of acid and water is usually prepared in a tank. The steel work piece is immersed in the acid solution, the acid reacts with the impurities and picks away the impurities, leaving it clean.
2. **DRY BLAST CLEANING:** is also known as abrasive blasting is a mechanical cleaning

process used to remove impurities, scale and rust from steel surfaces. A suitable blasting media (e.g Sand), blasting machine, hoses and nozzles are employed. compressed air propels, the blasting media onto the steel surface, forcibly removing impurities, scale and rust.

3. WET BLAST CLEANING: It is also known as slurry blasting or hydra blasting. It is a mechanical cleaning process that uses a mixture of water and abrasive media to remove impurities, scale, rust from steel surface.

CATHODIC PROTECTION

When different metals and alloys are joined together in an electrolyte, one of them generally corrodes at a higher rate while the other is protected. Steel generally corrodes at a greater rate under these circumstances but when joined to certain metals, notably aluminium, magnesium and zinc, steel is protected, and the coupled metal corrodes at an increased rate. In practice, a large steel structure can be protected by attaching blocks of such metals to it. The steel and the anode materials must, of course, be immersed in an electrolyte of suitable conductivity. Similar results can be obtained by impressing a current so that the steel potential is less than 0.56 V compared with a standard hydrogen electrode. Cathodic protection (CP) is a technique used to prevent corrosion on metal surfaces, particularly in harsh environment like seawater or soil.

SACRIFICIAL (GALVANIC) ANODE CATHODIC PROTECTION SYSTEM;

Uses a more active material (anode) to protect the steel pipeline. Anodes are attached to pipeline and provides a current to protect the pipeline. No external power source is needed, suitable for submerged pipelines.

IMPRESSED CURRENT CATHODIC PROTECTION SYSTEM;

Uses external power source to drive a current to protect the pipeline. It can vary current output to meet system requirements.

ANODE MATERIALS

Anode, materials, such as aluminum, zinc, magnesium and their alloys are used to protect steel structures in seawater by corroding preferentially to the steel. The key properties of anode materials are;

- i. Current capacity (How much electric charge it can supply over time (ampere-hour) rating)
- ii. Operating voltage/driving voltage (pushing power that drives the electric current from the anode to the metal to be protected)

Anodes are made out of alloys that are more electrochemically active than the material they are intended to protect from corrosion. As a result of interactions with surrounding structures and fluids, the anode depletes during the lifetime of the designed structure. The anodes that are used to prevent corrosion in cathodic design systems are called sacrificial anodes. The principle of the sacrificial anode is the same as the principle of cathodic protection, wherein metal surfaces come into contact with electrolytes, leading to an electrochemical reaction called corrosion. Metal in seawater is an example of this principle, as exemplified by a steel structure coming into contact with electrolytes in the ocean. Under normal circumstances, the steel will react with the

electrolytes and begin to corrode, growing structurally weaker and disintegrating over time. However, if the electrolyte contains another metal that is more active than the steel, then the more active metal will corrode preferentially, thus protecting the steel from corrosion.

As described above, these more active metals are the sacrificial anodes positioned in the same electrolyte in order to corrode preferentially compared to the structure of the interest. The sacrificial anodes that are used in offshore cathodic protection are intended to be consumed over the lifetime of the protected structure. For example, in cathodic protection designs where anodes are buried to protect land structures, a special backfill material surrounds the anode in order to insure that the anode produces the desired output. The backfill material acts as regulator of uniform, constant, well-distributed current output.

Finally, the sacrificial anode works by introducing another metal surface that is a more electronegative and anodic surface. The system allows the current to flow from the newly introduced metal, as an anode, and the protected. In relation to standard carbon steel, which is used for the construction of structures and piping components placed in the seawater, materials such as aluminum, zinc, magnesium, and their purpose-designed alloys are more active, and hence, they corrode preferentially to the steel structure, protecting it.

ANODE MANUFACTURING

The manufacturing process for anodes is a structured activity in which the following are detailed in the procedure.

- The quality and source of the raw material
- The melting and casting practices, including temperature and mold used

ANODE STEEL INSERTS

The inserts in the anode are the steel bodies cast into the anode material. These inserts are often made out of a ductile grade of steel that can be welded to the structure to attach the anode.

CONNECTIONS TO THE ANODE

A low resistance, mechanically adequate attachment is required for good protection and resistance to mechanical damage. In the process of providing electrons for the cathodic protection of a less active metal, the more active metal corrodes. The more active metal (anode) is sacrificed to protect the less active metal (cathode). The amount of corrosion depends on the metal being used as an anode, but it is directly proportional to the amount of current supplied.

Sacrificial anodes are normally supplied with lead wires or cast-in straps to facilitate their connection to the structure being protected. The lead wires may be attached to the structure by welding or mechanical connections.

Example 1;

A ship hull is a complex shape root amendable to simple surface area calculation. It is normal to use the formula

$$A = (1.8 \times LBP \times D) + (BC \times LBP \times B)$$

Where,

A = wetted surface area

LBP = Length between perpendiculars

D = Draft

BC = Block coefficient

B = Breadth

Typical block coefficient for various classifications of vessel are:

Naval vessels 0.55

Trawlers 0.55

Passenger vessels 0.60

Cargo vessels 0.75

Tankers 0.8-0.9

ANODE UTILIZATION FACTOR

Bracelet type/short flush $0.315f/\sqrt{A}$ 0.80

mounted

Long flush mounted R/2.5 0.85

Long and slender stand - off 0.90

Short and slender stand off 0.85

A cargo ship with the following dimensions requires cathodic protection

- Length between perpendicular (LBP) = 150m

- Breadth (B) = 20m

- Draft (T) = 8m

- Block coefficient (BC) = 0.75

- Steel current density = -0.1 Am/m^2

- Steel potential = -0.7 v

- Coating breakdown = 1

Assuming a long flush mounted magnesium anode with the following properties is employed.

Anode dimension 0.5m by 0.13m

Anode potential = -1.7 v

Utilization factor = 0.85, mass of 14.5kg

Resistivity of environment = $250 \text{ ohms/cm} = 2.5 \text{ ohms/m}$

Anode current capacity = 1100 A.h/kg

- Estimate the wetted surface area of the ship hull that requires cathodic protection.
- Estimate the number of anode required to protect the ship
- Determine if the number of anode will be suitable for the ships protection for 40 years

Solution

Current demand (I_d)

$$I_d = A_S \times I_C \times F_C$$

$$A_S = (1.8 \times LBP \times D) + (BC \times LBP \times B)$$

$$= (1.8 \times 150 \times 8) + (0.75 \times 150 \times 20)$$

$$= 2160 + 2250$$

$$= 4410 \text{m}^2$$

$$I_d = 4410 \times 0.1 \times 10^{-3} \times 1$$

$$= 4410 \times 0.0001 \times 1$$

$$= 0.441 \text{A}$$

Anode Resistivity

$$R_a = R/2 \times S$$

$$0.5 + 0.13 \times 0.63$$

$$S = 2$$

$$= 2$$

$$= 2$$

$$= 0.315$$

$$2.5 \times 2.5$$

$$= 2(0.315) \times 0.63 = 3.968 \Omega \text{m}$$

Driving Voltage

$$E_S - E_A = -0.7 - (-1.7)$$

$$= 1$$

Anode Current

$$I_a = E_S - E_A / R_a$$

$$-0.7 - (-1.7) / 1$$

$$= 3.968$$

$$= 3.968$$

$$= 0.252 \text{A}$$

$$= 0.252 \text{A}$$

Number of anode required

$$0.441$$

$$I_d / I_a = 0.252$$

$$= 1.75$$

$$\approx 2 \text{ anodes.}$$

Expected years of the configuration

$$M_a N_a E_c U_f \geq 8760 \times T_d I_c$$

M_a = Mass of anode

N_a = Number of anode

E_c = Current capacity of anode

U_f = Utilization factor

T_d = Service life span in years

I_c = Current density

$$= 14.5 \times 2 \times 1100 \times 0.85 \geq \times 8760 \times 10 \times 0.0001$$

$$= 27115 \geq 8.76$$

□ It is suitable

Example 2;

.An offshore pipeline with length of 2000m and a diameter of 0.254m requires sacrificial anode protection. The pipeline steel has a current density of 0.1mA/m^2 , and the resistivity of the environment is $400\Omega\text{cm}$. a coating factor of 10% is applied. A magnesium bracelet anode with dimensions 0.5m long and 0.13m in diameter are used. The anode potential is -0.75v and the average potential of the pipeline is -0.7v .

Calculate;

1. The total current required to protect the pipeline
2. The number of anodes required to provide the necessary protection.

Solution

Current demand by pipeline

$$I_d = A \times F_c \times I_c$$

$$A = \pi dL$$

$$22$$

$$= 7 \times 0.254 \times 2000$$

$$= 1596.57\text{m}^2$$

$$I_d = 1596.57 \times 0.1 \times 0.1 \times 10^{-3}$$

$$= 0.01596557$$

$$\approx 0.05966$$

Anode Current

$$0.315 \times \diamond$$

$$R_n = \sqrt{\diamond \diamond}$$

$$0.315 \times 4$$

$$= \sqrt{\diamond \diamond}$$

$$(400\Omega\text{cm} = 4\Omega\text{m})$$

$$A_a = \pi DL$$

$$= 3.143 \times 0.13 \times 0.5$$

$$= 0.2043$$

Anode resistivity

$$0.315 \times 4$$

$$R_a = 0.2043$$

$$1.26$$

$$= 0.452$$

$$= 2.788 \Omega m$$

Anode current

$$I_a = E_s - E_A / R_a \text{ (driving voltage)}$$

$$-0.7 - (-1.75) / 1.05$$

$$= 2.788$$

$$= 2.788$$

$$= 0.3766 A$$

Number of anode required

$$0.15966$$

$$I_d / I_a = 0.3766$$

$$= 0.424$$

$$\approx 1 \text{ anode}$$